Chapter 8. Generating Functions

A Function Is Worth Many Numbers

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A generating function is "a clothesline on which we hang all elements of a sequence" (H. Wilf): one function stores infinitely many numbers. Operations on the function then correspond to combinatorial constructions, which makes generating functions the most powerful tool of enumeration.

8.1 Ordinary generating functions

8.1.1 Recurrence relations and generating functions

Motivation. A recurrence $a_{n+1} = 4a_n - 100$ is infinitely many equations. Multiplying the n-th by x^{n+1} and summing packs them into one equation for a single power series. Solving it and reading off coefficients by partial fractions gives an explicit formula.

A lake's frog population quadruples each year and 100 frogs are removed at the start of each year: $a_0 = 50$ and

$$a_{n+1} = 4a_n - 100 \qquad (n \geq 0). \tag{8.1}$$

Computing a_{87} by iterating (8.1) wastes 86 steps; we want an explicit formula.

Definition 8.1 (Ordinary generating function)

For a sequence $(f_n)_{n\geq 0}$ of reals, the **formal power series** $F(x) = \sum_{n\geq 0} f_n x^n$ is its **ordinary generating function**.

A formal power series is determined by its coefficients and need not converge (e.g. $\sum n!x^n$); two series are equal iff all coefficients agree.

Multiply (8.1) by x^{n+1} and sum over $n \geq 0$. With $G(x) = \sum a_n x^n$ and $\sum_{n\geq 0} x^n = \frac{1}{1-x}$:

$$G(x) - a_0 = 4xG(x) - \frac{100x}{1-x}, \qquad G(x) = \frac{50}{1-4x} - \frac{100x}{(1-x)(1-4x)}. \tag{8.2}$$

Partial fractions give $\frac{100x}{(1-x)(1-4x)} = \frac{100}{3}\left(\frac{1}{1-4x} - \frac{1}{1-x}\right)$. Since $\frac{1}{1-4x} = \sum 4^n x^n$,

$$a_n = 50 \cdot 4^n - 100 \cdot \frac{4^n - 1}{3}, \tag{8.3}$$

and one checks directly that (8.3) satisfies $a_0 = 50$ and (8.1).

The method. (1) Define $G(x) = \sum a_n x^n$. (2) Turn the recurrence into an equation for G (multiply by x^n, x^{n+1} or x^{n+k} and sum over $n \geq 0$). (3) Solve for G. (4) Read off the coefficient of x^n.

Example 8.2

\$1000 earns 5% interest per year and \$500 is deposited each year: $a_0 = 1000$, $a_{n+1} = 1.05a_n + 500$.

Solution. As above, $G(x) - 1000 = 1.05xG(x) + \frac{500x}{1-x}$, so

$$G(x) = \frac{1000}{1-1.05x} + \frac{500x}{(1-x)(1-1.05x)} = \frac{1000}{1-1.05x} + \frac{10000}{1-1.05x} - \frac{10000}{1-x},$$

and $a_n = 11000 \cdot 1.05^n - 10000$.

Example 8.3

$a_0 = 0$, $a_1 = 1$, $a_{n+2} = 3a_{n+1} - 2a_n$. Multiplying by x^{n+2} and summing: $G - x = 3xG - 2x^2G$. So

$$G(x) = \frac{x}{1 - 3x + 2x^2} = \frac{1}{1 - 2x} - \frac{1}{1 - x}, \qquad a_n = 2^n - 1.$$

8.1.2 Products of generating functions

Motivation. Multiplying generating functions convolves their coefficients. Combinatorially this means cutting $[n]$ into an initial and a final interval and building independent structures on the two pieces. Many counts that look like awkward sums become simple products.

Lemma 8.4

If $A(x) = \sum a_n x^n$, $B(x) = \sum b_n x^n$ and $c_n = \sum_{i=0}^{n} a_i b_{n-i}$, then $A(x)B(x) = C(x) := \sum c_n x^n$.

Proof. Multiplying out, every product $a_i x^i \cdot b_j x^j$ appears once, and it contributes to x^n iff $j = n - i$. ■

Theorem 8.5 (Product formula)

Let a_n (b_n) be the number of structures of a first (second) kind on an n-element set. Let c_n be the number of ways to cut $[n]$ into the intervals $S = \{1, \dots, i\}$ and $T = \{i + 1, \dots, n\}$ (possibly empty) and build a structure of the first kind on S and of the second kind on T. Then $A(x)B(x) = C(x)$.

Proof. For a fixed cut point i there are $a_i b_{n-i}$ choices, so $c_n = \sum_{i=0}^{n} a_i b_{n-i}$; apply Lemma 8.4. ■

By induction, the same holds for any fixed number of intervals.

Example 8.6

An n-day semester is split into a theoretical part (first k days) and a lab part ($n - k$ days); one holiday is chosen in the first part and two in the second. The number of plans is $f_n = \binom{n+1}{4}$.

Solution. $f_n = \sum_k k\binom{n-k}{2}$. The pieces have generating functions $A(x) = \sum_{k \geq 1} kx^k = \frac{x}{(1-x)^2}$ (differentiate $\sum x^k$) and $B(x) = \sum_{m \geq 2} \binom{m}{2} x^m = \frac{x^2}{(1-x)^3}$. So

$$F(x) = A(x)B(x) = \frac{x^3}{(1 - x)^5} = x^3 \sum_{n \geq 0} \binom{n + 4}{4} x^n = \sum_{n \geq 3} \binom{n + 1}{4} x^n.$$

Here $(1 - x)^{-5} = \sum_n \binom{n+4}{4} x^n$, by Theorem 4.15 or because $\binom{n+4}{4}$ counts the multisets of size n from $[5]$ (Theorem 3.21).

Example 8.7

If instead any set of independent-study days is chosen in each part, each part has generating function $C(x) = \sum 2^k x^k = \frac{1}{1-2x}$. So $F(x) = \frac{1}{(1-2x)^2} = \frac{1}{2}C'(x) = \sum_n (n + 1)2^n x^n$, and $g_n = (n + 1)2^n$.

Example 8.8

Split an n-day semester into three parts and choose any number of holidays in the first, an odd number in the second and an even number in the third. An m-set has 2^{m-1} odd and 2^{m-1} even subsets for $m \geq 1$ (by Theorem 4.2). So the generating functions are $\frac{1}{1-2x}$, $\frac{x}{1-2x}$ and $1 + \frac{x}{1-2x} = \frac{1-x}{1-2x}$, and

$$G(x) = \frac{x(1-x)}{(1-2x)^3} = -\frac{1}{4} \cdot \frac{1}{1-2x} + \frac{1}{4} \cdot \frac{1}{(1-2x)^3}.$$

With $(1-2x)^{-3} = \sum_n \binom{n+2}{2} 2^n x^n$ this gives $g_n = \frac{1}{4}(\binom{n+2}{2} - 1)2^n = 2^{n-3}n(n+3)$.

Example 8.9

With $p_{\leq k}(n)$ the number of partitions of n into parts of size at most k,

$$\sum_{n \geq 0} p_{\leq k}(n) x^n = \prod_{i=1}^{k} \frac{1}{1-x^i} = (1+x+x^2+\cdots)(1+x^2+x^4+\cdots)\cdots(1+x^k+x^{2k}+\cdots). \quad (8.4)$$

Solution. A term x^n of the product picks x^{ij_i} from the i-th factor with $1j_1+2j_2+\cdots+kj_k = n$. This corresponds to the partition of n with j_i parts equal to i, and this correspondence is bijective. By Theorem 5.17 this is also the generating function of partitions into at most k parts.

Example 8.10

For all partitions,

$$\sum_{n \geq 0} p(n) x^n = \prod_{k \geq 1} \frac{1}{1-x^k}, \quad (8.5)$$

by the same argument. The infinite product is well defined: only the first n factors affect the coefficient of x^n.

Example 8.11 (Euler)

The number $p_{\text{odd}}(n)$ of partitions of n into odd parts equals the number $p_d(n)$ of partitions into distinct parts.

Solution. It suffices to compare generating functions:

$$\sum_n p_{\text{odd}}(n) x^n = \prod_{i \text{ odd}} \frac{1}{1-x^i}, \qquad \sum_n p_d(n) x^n = \prod_{i \geq 1} (1+x^i) = \prod_{i \geq 1} \frac{1-x^{2i}}{1-x^i}.$$

In the last product every numerator $1-x^{2i}$ cancels the denominator factor $1-x^{2i}$, leaving $\prod_{i \text{ odd}} \frac{1}{1-x^i}$.

8.1.2.1 The Catalan numbers

A jar starts empty; each day a coin is put in or taken out, never going negative, and after $2n$ days the jar is empty again. Let c_n ($c_0 = 1$) count such histories: sequences $b_1, \ldots, b_{2n} \in \{\pm 1\}$ with $\sum_j b_j = 0$ and all partial sums $\sum_{j \leq k} b_j \geq 0$ (**good sequences**).

Decomposition. For $n > 0$ let day $2i$ be the first day after the start with an empty jar (the jar

can only be empty after an even number of days). Then days $2i+1,\dots,2n$ form a good sequence of length $2(n-i)$. Days $1,\dots,2i$ form a **very good** sequence, one whose partial sums are positive strictly inside. Stripping its first $+1$ and last -1 gives a good sequence of length $2(i-1)$, bijectively. So very good sequences have generating function $\sum_{i\ge 1} c_{i-1}x^i = xC(x)$, and the product formula gives

$$C(x) - 1 = xC(x)\cdot C(x), \qquad \text{i.e.} \quad xC(x)^2 - C(x) + 1 = 0. \tag{8.6}$$

Of the roots $\frac{1\pm\sqrt{1-4x}}{2x}$ only the minus sign gives a power series with constant term 1 (the plus sign has a pole at 0), so

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}. \tag{8.7}$$

By Theorem 4.15, $\sqrt{1-4x} = \sum_n \binom{1/2}{n}(-4x)^n = 1 - 2\sum_{n\ge 1}\frac{1}{n}\binom{2n-2}{n-1}x^n$ (the same computation as in Example 4.16). Hence

$$C(x) = \sum_{n\ge 0}\frac{1}{n+1}\binom{2n}{n}x^n, \qquad c_n = \frac{1}{n+1}\binom{2n}{n}.$$

These are the **Catalan numbers** $1,1,2,5,14,42,\dots$, which count well over 150 kinds of objects (see Chapter 14).

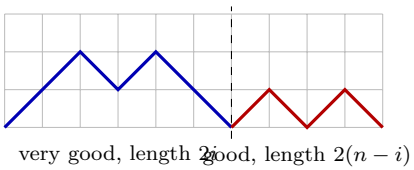


Figure. *A good sequence as a lattice path ($+1$ up, -1 down) that never goes below the axis. Cutting at the first return to the axis gives the decomposition $C - 1 = xC\cdot C$.*

8.1.3 Compositions of generating functions

Motivation. Cutting $[n]$ into an unspecified number of nonempty intervals, each carrying a structure, corresponds to substituting one generating function into another. This is the composition of formal power series, defined when the inner series has no constant term.

For $F(x) = \frac{1}{1-x}$ we would like $F(G(x)) = 1 + G + G^2 + \cdots$. This sum of series is well defined iff each coefficient receives only finitely many contributions, which holds when G has constant term 0: then G^m is divisible by x^m.

Definition 8.12 (Composition)

For a formal power series $F(x) = \sum f_n x^n$ and G with constant term 0, $F(G(x)) := \sum_{n\ge 0} f_n G(x)^n$.

Theorem 8.13

Let a_n count structures of some kind on an n-set, with $a_0 = 0$. Let h_n ($h_0 = 1$) count the ways to cut $[n]$ into any number of nonempty intervals and put such a structure on each. Then $H(x) = \frac{1}{1-A(x)}$.

Proof. By the product formula (for k factors), $A(x)^k$ counts the cuts into exactly k intervals, each with a structure; nonempty intervals are automatic since $a_0 = 0$. As $a_0 = 0$, A^k has no

terms below x^k, so the sum over k is well defined, and with the constant term $h_0 = 1$ we get $H = 1 + \sum_{k\geq 1} A^k = \frac{1}{1-A}$. ■

Empty intervals must be excluded, since otherwise there would be infinitely many cuts.

Example 8.14

n soldiers in a line are split into nonempty units and a commander is named in each. The number of ways is $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$, with $\alpha, \beta = \frac{3\pm\sqrt{5}}{2}$.
Solution. Here $a_k = k$, $A(x) = \frac{x}{(1-x)^2}$, so $H(x) = \frac{1}{1-A(x)} = 1 + \frac{x}{1-3x+x^2}$. The roots of $x^2 - 3x + 1$ are α, β with $\alpha\beta = 1$, and partial fractions give

$$\frac{1}{1-3x+x^2} = \frac{1}{\sqrt{5}}\Big(\frac{\alpha}{1-\alpha x} - \frac{\beta}{1-\beta x}\Big) = \frac{1}{\sqrt{5}}\sum_n(\alpha^{n+1} - \beta^{n+1})x^n.$$

So the coefficient of x^n in H is $\frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$: $1, 3, 8, 21, 55, \ldots$.

Theorem 8.15 (Compositional formula)

Let a_n count structures of a first kind on an n-set ($a_0 = 0$) and b_n structures of a second kind ($b_0 = 1$). Let g_n ($g_0 = 1$) count the ways to cut $[n]$ into nonempty intervals, put a first-kind structure on each interval and a second-kind structure on the set of intervals. Then $G(x) = B(A(x))$.

Proof. With exactly k intervals there are b_k structures on the set of intervals, and the generating function of the interval structures is $A(x)^k$ (product formula). So this case contributes $b_k A(x)^k$, and $G = \sum_k b_k A(x)^k = B(A(x))$. ■

Example 8.16

The officer splits the line into nonempty units and chooses any subset of the units for night duty. Here $a_k = 1$ ($k \geq 1$) and $b_m = 2^m$, so $A = \frac{x}{1-x}$, $B = \frac{1}{1-2x}$ and

$$G(x) = \frac{1}{1 - \frac{2x}{1-x}} = \frac{1-x}{1-3x} = 1 + \sum_{n\geq 1} 2 \cdot 3^{n-1}x^n.$$

For $n \geq 1$ there are $2 \cdot 3^{n-1}$ ways.

8.2 Exponential generating functions

8.2.1 Recurrence relations and exponential generating functions

Motivation. Sequences that grow like $n!$ have ordinary generating functions with no closed form. Dividing the n-th term by $n!$ tames them, and the factor $(n + 1)$ that typically appears in such recurrences then disappears.

Example 8.17

$a_0 = 1$, $a_{n+1} = (n + 1)(a_n - n + 1)$ for $n \geq 0$. Then $a_n = n! + n$.

Definition 8.18 (Exponential generating function)

For a sequence $(f_n)_{n\geq 0}$, the formal power series $F(x) = \sum_{n\geq 0} f_n \frac{x^n}{n!}$ is its **exponential generating function** (the constant sequence 1 has e^x).

Solution of Example 8.17. Multiply the recurrence by $\frac{x^{n+1}}{(n+1)!}$ and sum over $n \geq 0$:

$$\sum_{n\geq 0} a_{n+1}\frac{x^{n+1}}{(n+1)!} = \sum_{n\geq 0} a_n \frac{x^{n+1}}{n!} - \sum_{n\geq 0}(n-1)\frac{x^{n+1}}{n!}, \tag{8.8}$$

i.e. $A(x) - 1 = xA(x) - x^2e^x + xe^x$. Hence

$$A(x) = \frac{1}{1-x} + xe^x = \sum_{n\geq 0} n!\frac{x^n}{n!} + \sum_{n\geq 1} n\frac{x^n}{n!},$$

since the equation reads $A(x)(1-x) = 1 + xe^x(1-x)$; hence $a_n = n! + n$.

Example 8.19

$f_0 = 0$, $f_{n+1} = 2(n+1)f_n + (n+1)!$. Multiplying by $\frac{x^{n+1}}{(n+1)!}$ and summing: $F(x) = 2xF(x) + \frac{x}{1-x}$. So $F(x) = \frac{x}{(1-x)(1-2x)} = \sum_n (2^n - 1)x^n$, and $f_n = (2^n - 1)\, n!$.

8.2.2 Products of exponential generating functions

Motivation. Exponential generating functions multiply by a binomial convolution. Combinatorially, this splits $[n]$ into two *arbitrary* subsets rather than two intervals, the natural operation for labelled structures.

Lemma 8.20

For exponential generating functions $A(x) = \sum a_i \frac{x^i}{i!}$, $B(x) = \sum b_k \frac{x^k}{k!}$ and $c_n = \sum_{i=0}^{n} \binom{n}{i} a_i b_{n-i}$, we have $A(x)B(x) = C(x) := \sum c_n \frac{x^n}{n!}$.

Proof. A general term of the product is $a_i \frac{x^i}{i!} \cdot b_j \frac{x^j}{j!} = a_i b_j \binom{i+j}{i} \frac{x^{i+j}}{(i+j)!}$, and it contributes to $\frac{x^n}{n!}$ iff $i + j = n$. ∎

Theorem 8.21 (Product formula, exponential version)

Let a_n, b_n count structures of two kinds on n-sets, and let c_n count the ways to split $[n]$ into disjoint subsets $S \cup T = [n]$ (no other restriction) and build a first-kind structure on S and a second-kind structure on T. Then $A(x)B(x) = C(x)$ for the exponential generating functions.

Proof. If $|S| = i$, there are $\binom{n}{i}$ choices of S, then a_i and b_{n-i} structures; so $c_n = \sum_i \binom{n}{i} a_i b_{n-i}$, and Lemma 8.20 applies. ∎

Theorem 8.5 cuts a *linearly ordered* set into intervals; Theorem 8.21 splits a set into arbitrary subsets.

Example 8.22

A coach splits n players into two groups, lines up each group, and gives everyone in the first group an orange, white or blue shirt. Here $a_k = k!\,3^k$, so $A(x) = \frac{1}{1-3x}$, and $b_m = m!$, so $B(x) = \frac{1}{1-x}$. Then

$$C(x) = \frac{1}{(1-3x)(1-x)} = \frac{3}{2}\cdot\frac{1}{1-3x} - \frac{1}{2}\cdot\frac{1}{1-x} = \sum_n \frac{3^{n+1}-1}{2}x^n,$$

and $c_n = n!\,\frac{3^{n+1}-1}{2}$.

Example 8.23

For fixed k, the exponential generating function of the Stirling numbers of the second kind is

$$S_k(x) := \sum_{n\geq k} S(n,k)\frac{x^n}{n!} = \frac{1}{k!}(e^x-1)^k. \tag{8.9}$$

Solution. A nonempty block carries "no structure" in exactly one way, with exponential generating function $\sum_{m\geq 1}\frac{x^m}{m!} = e^x - 1$. By the product formula, $(e^x-1)^k$ counts partitions into k *ordered* blocks; each unordered partition arises $k!$ times. Expanding $(e^x-1)^k = \sum_i \binom{k}{i}(-1)^i e^{(k-i)x}$ recovers Theorem 7.5.

Differentiation shifts exponential generating functions: $\left(\sum a_n\frac{x^n}{n!}\right)' = \sum a_{n+1}\frac{x^n}{n!}$.

Example 8.24

The exponential generating function of the Bell numbers is $B(x) = e^{e^x-1}$.
Solution. Multiply $B(n+1) = \sum_i \binom{n}{i}B(i)$ (Theorem 5.12) by $\frac{x^n}{n!}$ and sum. The left side is $B'(x)$, and by Lemma 8.20 the right side is $B(x)e^x$. So $(\ln B)' = e^x$, $\ln B(x) = e^x + C$, and $B(0) = 1$ forces $C = -1$.

8.2.3 Compositions of exponential generating functions

Motivation. Partitioning a labelled set into blocks, each carrying a structure, corresponds to exponentiating the generating function of the structure. This exponential formula explains at once why permutations (sets of cycles) are counted by $e^{\ln\frac{1}{1-x}} = \frac{1}{1-x}$.

Theorem 8.25 (Exponential formula)

Let a_n count structures on n-sets ($a_0 = 0$) and let h_n ($h_0 = 1$) count the ways to partition $[n]$ into any number of nonempty blocks and put a structure on each block. Then $H(x) = e^{A(x)}$ for the exponential generating functions.

Proof. By Theorem 8.21 (for k factors), $A(x)^k$ counts the decompositions of $[n]$ into k *ordered* blocks with structures, so $A(x)^k/k!$ counts those into exactly k unordered blocks. As $a_0 = 0$, A^k has no terms below x^k, so $H = 1 + \sum_{k\geq 1}\frac{A^k}{k!} = e^{A(x)}$. ∎

Integrating $\frac{1}{1-x} = \sum x^n$ termwise (the formal integral of $\sum a_n x^n$ is $\sum a_n\frac{x^{n+1}}{n+1}$) gives

$$\sum_{k\geq 1}\frac{x^k}{k} = \ln\frac{1}{1-x}.$$

Example 8.26

n people form groups and each group sits at a circular table. With $a_k = (k-1)!$ (Example 6.10), $A(x) = \sum_k \frac{x^k}{k} = \ln \frac{1}{1-x}$, so $H(x) = \frac{1}{1-x} = \sum n! \frac{x^n}{n!}$ and $h_n = n!$. Combinatorially: a permutation is exactly a set of cycles.

Example 8.27

The partitions of $[n]$ into blocks of sizes 3, 4 and 9 have exponential generating function $e^{x^3/3! + x^4/4! + x^9/9!}$. Blocks of size 3 only give $e^{x^3/3!}$ (exponential formula with $T(x) = x^3/3!$), and similarly for 4 and 9; the three kinds are combined by the product formula.

Theorem 8.28 (Compositional formula, exponential version)

Let a_n count first-kind structures ($a_0 = 0$) and b_n second-kind structures ($b_0 = 1$). Let g_n count the ways to partition $[n]$ into nonempty blocks, put a first-kind structure on each block and a second-kind structure on the set of blocks. Then $G(x) = B(A(x))$.

Proof. With k blocks there are b_k structures on the set of blocks, and by Theorem 8.21 the blocks with their structures contribute $\frac{A(x)^k}{k!}$. So $G = \sum_k b_k \frac{A(x)^k}{k!} = B(A(x))$; the sum is defined since $a_0 = 0$. ■

Example 8.29

n distinct cards are split into nonempty subsets of even size, the cards in each subset are ordered, and the subsets are put in a line. Here $a_n = n!$ for even $n \geq 2$ and 0 otherwise, so $A(x) = \sum_{n \geq 2 \text{ even}} x^n = \frac{x^2}{1-x^2}$; and $b_n = n!$, so $B(x) = \frac{1}{1-x}$. Then

$$G(x) = \frac{1}{1 - \frac{x^2}{1-x^2}} = \frac{1-x^2}{1-2x^2} = 1 + \sum_{m \geq 0} 2^m x^{2m+2},$$

so $g_n = 0$ for odd n and $g_n = 2^{n/2-1}\, n!$ for even $n \geq 2$.